

Name:

In-class final, EC524/424, Winter 2026

120 points possible

Instructions A short response and/or sketch suffices for most of the following questions. Do not write more than necessary. Excessively long responses will be penalized.

1. (4 points) Briefly explain how prediction fundamentally differs from causal inference.

2. (4 points) Explain the difference between supervised and unsupervised learning.

3. (4 points) When we say one model is more *flexible* or less *parametric* than another, what do we mean? Give one example of a relatively flexible model and one example of a relatively less flexible model.

4. (4 points) Define *irreducible error*. Why can no learning algorithm eliminate it completely?

5. (4 points) In the context of the bias-variance tradeoff, define what we mean by *bias*.

6. (4 points) In the context of the bias-variance tradeoff, define what we mean by *variance*.

7. (4 points) Describe how increasing a model's flexibility typically affects training error and test error.

8. (4 points) Why do we generally prefer k -fold cross-validation over the validation-set approach?

9. (4 points) What does it mean to *tune* a model?

10. (4 points) What is bootstrapping, and why is it useful in prediction problems?

11. (4 points) Why might we prefer shrinkage methods despite their potential for bias?

12. (4 points) Why do we normalize our predictors before fitting a penalized regression model?

13. (4 points) What is the key difference between ridge and lasso? Which one can set coefficients exactly equal to zero?

14. (4 points) Explain how k -nearest neighbors (KNN) makes a prediction for a new observation.

15. (4 points) Why might we prefer logistic regression over linear regression for classification tasks?

16. (4 points) Suppose you raise a classifier's probability cutoff from 0.5 to 0.8. What would likely happen to sensitivity and precision?

17. (4 points) Draw a confusion matrix for a binary classifier and explain what information it provides.

18. (4 points) Define the *null classifier*. Why is it a useful benchmark, especially when classes are imbalanced?

19. (4 points) Describe, at a high level, how a decision tree is grown.

20. (4 points) Why do we prune decision trees?

21. (4 points) Why do classification trees usually split with Gini or entropy rather than accuracy?

22. (4 points) Explain how combining many weak or unstable learners can improve prediction. Give one example of an algorithm that uses this idea.

23. (4 points) What is the difference between bagging and boosting?

24. (4 points) How do random forests decorrelate trees?

25. (4 points) Explain the ROC curve. Specifically: What does it plot and what does it tell us?

26. (4 points) In an SVM, what makes an observation a *support vector*?

27. (4 points) What role does the *cost* (or penalty) parameter play in a soft-margin SVM?

28. (4 points) At a high level, how does a kernel allow an SVM to fit a nonlinear decision boundary?

29. (8 points) A classmate says, “K-means clustering is basically KNN without labels.”

a. In words, describe how K-means works.

b. Give one clear reason your classmate’s statement is wrong.